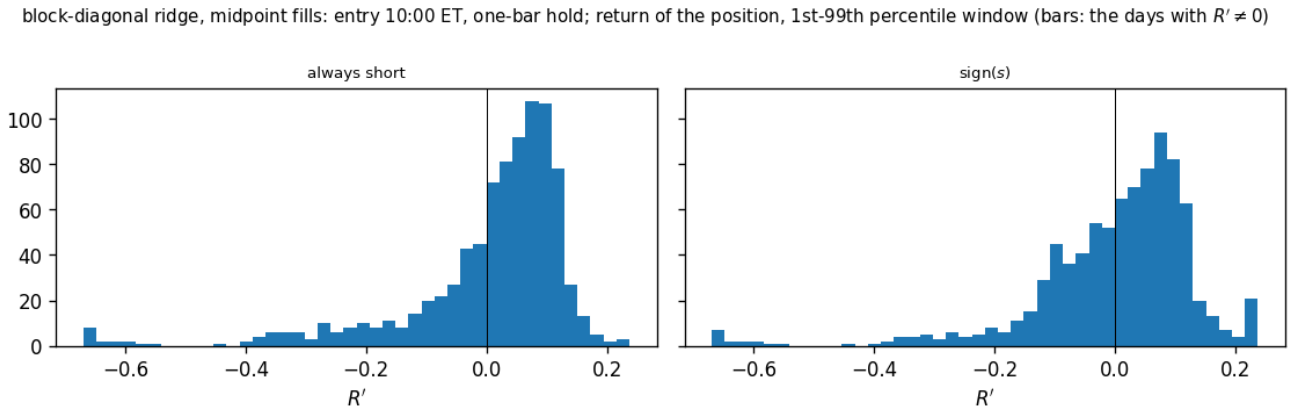
Rule table by strategy — entry 10:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	864	0.002	0.149	-1.12	0.25	-2.71	10.66	0.49	0.26	0	0.0	-3.74
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	864	-0.003	0.149	-1.12	0.67	-2.21	10.33	-0.50	-0.27	207	24.0	-4.27
block-diagonal ridge	864	0.001	0.149	-1.12	0.65	-2.15	10.51	0.14	0.07	219	25.3	-4.00
block-diagonal ridge, without the FOMC columns	864	0.001	0.149	-1.12	0.65	-2.15	10.50	0.10	0.06	221	25.6	-4.02
LightGBM†	864	-0.003	0.149	-1.12	0.67	-2.21	10.29	-0.62	-0.33	199	23.0	-4.39
XGBoost†	864	-0.004	0.149	-1.12	0.67	-2.19	10.25	-0.79	-0.43	190	22.0	-4.47
lasso (causally tuned)†	864	-0.004	0.149	-1.12	0.41	-2.29	10.26	-0.72	-0.39	205	23.7	-4.43
lasso ($\alpha = 10^{-4}$)	864	-0.001	0.149	-1.12	0.41	-2.30	10.39	-0.27	-0.15	208	24.1	-4.22
elastic net (causally tuned)†	864	-0.003	0.149	-1.12	0.41	-2.31	10.31	-0.52	-0.28	210	24.3	-4.34

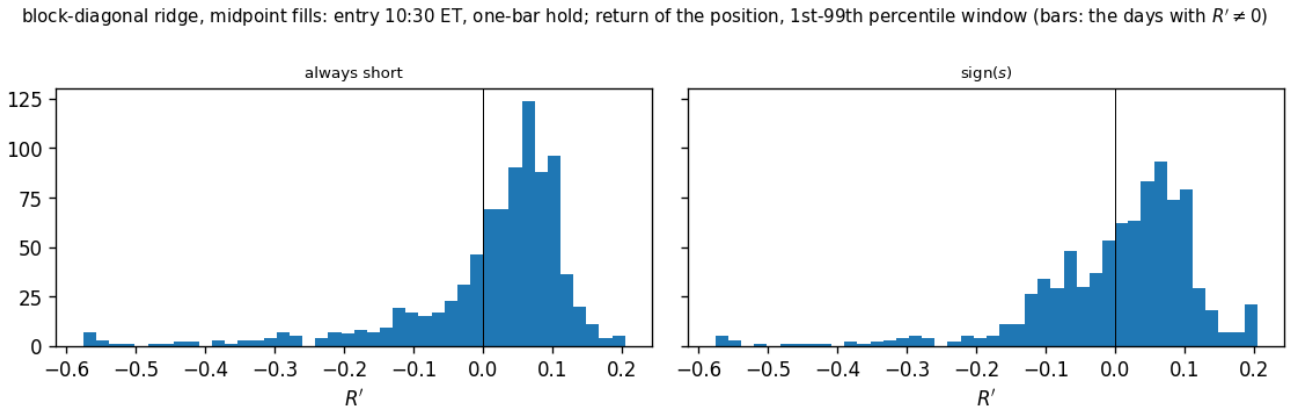


Distribution of the return of the position taken at 10:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 864 expiration days (the other 2 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 10:30 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.006	0.154	-1.96	0.21	-5.02	43.98	1.18	0.64	0	0.0	-2.44
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.005	0.154	-1.96	0.53	-4.57	43.65	0.88	0.48	199	23.0	-2.60
block-diagonal ridge	865	0.005	0.154	-1.96	1.19	-3.26	43.54	0.96	0.52	210	24.3	-2.59
block-diagonal ridge, without the FOMC columns	865	0.005	0.154	-1.96	1.19	-3.26	43.58	1.01	0.55	207	23.9	-2.55
LightGBM†	865	0.004	0.154	-1.96	0.83	-4.29	43.54	0.77	0.42	197	22.8	-2.66
XGBoost†	865	0.005	0.154	-1.96	0.83	-4.31	43.63	0.89	0.48	193	22.3	-2.60
lasso (causally tuned)†	865	0.004	0.154	-1.96	0.83	-4.33	43.52	0.74	0.40	201	23.2	-2.68
lasso ($\alpha = 10^{-4}$)	865	0.005	0.154	-1.96	1.19	-3.31	43.53	0.93	0.50	201	23.2	-2.60
elastic net (causally tuned)†	865	0.008	0.154	-1.96	1.19	-3.63	43.95	1.45	0.78	198	22.9	-2.32

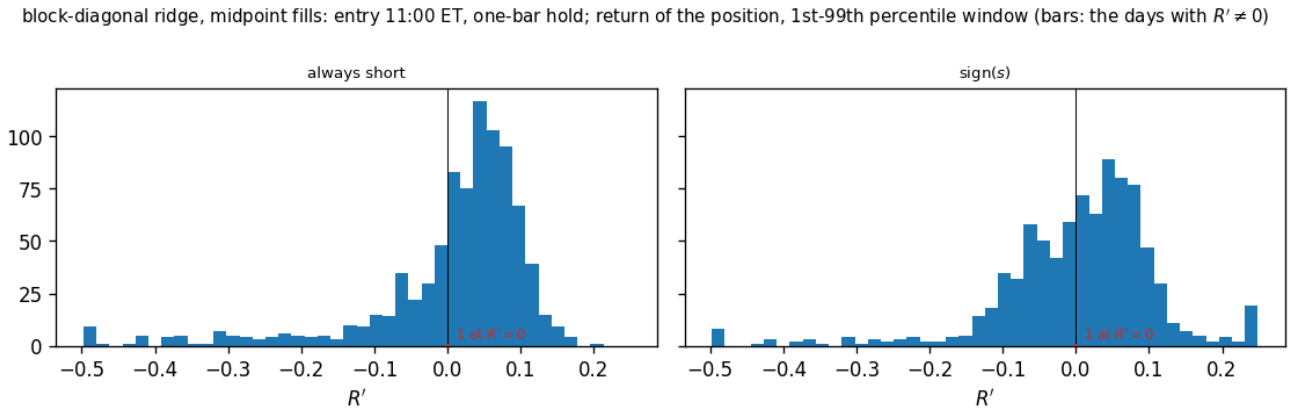


Distribution of the return of the position taken at 10:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 10:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.004	0.131	-1.27	0.21	-3.41	18.52	0.86	0.46	0	0.0	-3.21
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.010	0.130	-0.81	1.27	-0.21	18.38	2.33	1.26	269	31.1	-2.48
block-diagonal ridge	865	0.004	0.131	-1.27	0.67	-2.32	18.37	0.81	0.44	276	31.9	-3.28
block-diagonal ridge, without the FOMC columns	865	0.006	0.131	-0.81	1.27	-0.28	18.22	1.39	0.75	275	31.8	-2.98
LightGBM†	865	0.007	0.131	-0.81	1.27	-0.53	18.31	1.60	0.87	262	30.3	-2.87
XGBoost†	865	0.008	0.131	-0.81	1.27	-0.25	18.29	1.84	0.99	255	29.5	-2.74
lasso (causally tuned)†	865	0.006	0.131	-0.81	1.27	-0.48	18.24	1.28	0.69	273	31.6	-3.03
lasso ($\alpha = 10^{-4}$)	865	0.007	0.131	-0.81	1.27	-0.30	18.27	1.67	0.90	270	31.2	-2.83
elastic net (causally tuned)†	865	0.004	0.131	-1.27	0.67	-2.35	18.37	0.80	0.43	273	31.6	-3.27

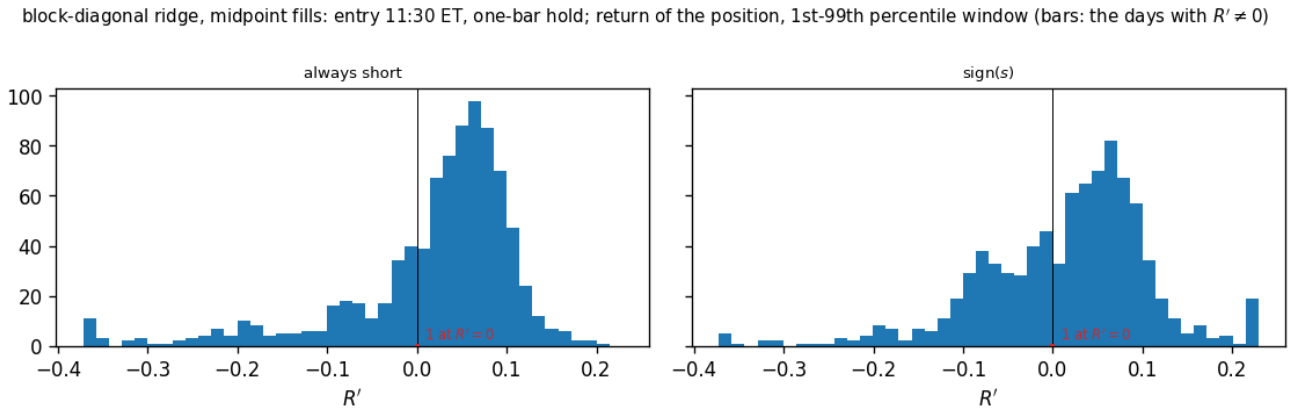


Distribution of the return of the position taken at 11:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 11:30 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	865	0.016	0.114	-1.06	0.21	-3.37	19.30	4.05	2.19	0	0.0	-2.13
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	865	0.019	0.114	-0.90	1.06	-0.16	17.81	4.80	2.59	231	26.7	-1.82
block-diagonal ridge	865	0.017	0.114	-0.65	1.06	1.13	16.89	4.49	2.42	252	29.1	-2.01
block-diagonal ridge, without the FOMC columns	865	0.017	0.114	-0.65	1.06	1.14	16.86	4.38	2.36	265	30.6	-2.07
LightGBM†	865	0.020	0.114	-0.65	1.06	1.06	17.08	5.11	2.76	221	25.5	-1.68
XGBoost†	865	0.020	0.114	-0.65	1.06	1.03	17.11	5.13	2.77	213	24.6	-1.67
lasso (causally tuned)†	865	0.018	0.114	-0.65	1.06	1.05	16.94	4.51	2.43	238	27.5	-1.99
lasso ($\alpha = 10^{-4}$)	865	0.016	0.114	-0.65	1.06	1.16	16.81	4.17	2.25	271	31.3	-2.17
elastic net (causally tuned)†	865	0.018	0.114	-0.65	1.06	1.05	16.95	4.54	2.45	236	27.3	-1.98

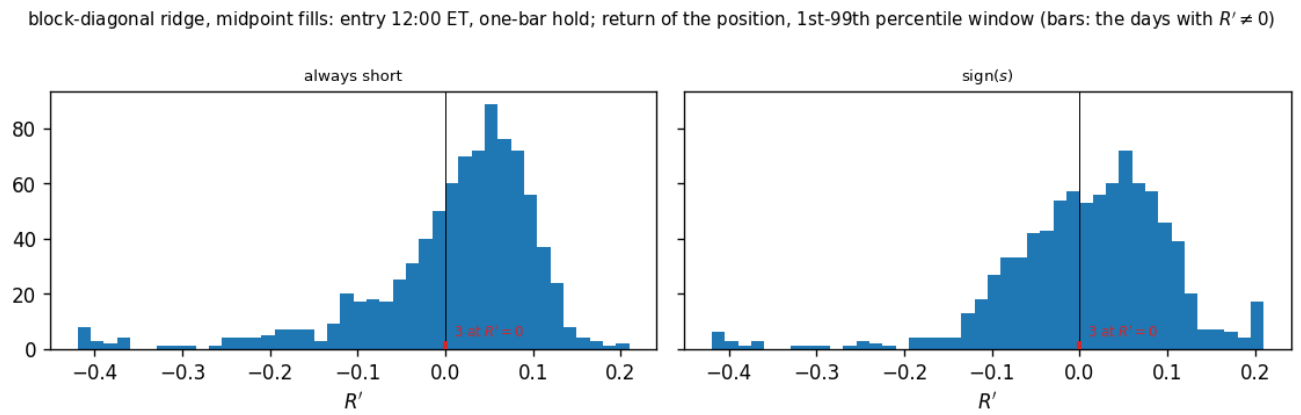


Distribution of the return of the position taken at 11:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 11:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 865 expiration days (the other 1 of the deck's 866 days have no quoted straddle at this stamp), 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.008	0.112	-1.02	0.22	-3.16	17.88	2.09	1.12	0	0.0	-3.42
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.006	0.112	-1.02	0.54	-2.53	17.42	1.52	0.82	259	29.9	-3.72
block-diagonal ridge	866	0.008	0.112	-1.02	0.54	-2.40	17.73	2.22	1.20	269	31.1	-3.36
block-diagonal ridge, without the FOMC columns	866	0.009	0.112	-1.02	0.54	-2.42	17.83	2.42	1.30	275	31.8	-3.26
LightGBM†	866	0.009	0.112	-1.02	0.54	-2.27	17.81	2.47	1.33	246	28.4	-3.26
XGBoost†	866	0.009	0.112	-1.02	0.54	-2.28	17.71	2.26	1.22	251	29.0	-3.35
lasso (causally tuned)†	866	0.005	0.112	-1.02	0.54	-2.52	17.34	1.34	0.72	265	30.6	-3.82
lasso ($\alpha = 10^{-4}$)	866	0.010	0.112	-1.02	0.54	-2.38	17.91	2.60	1.40	266	30.7	-3.16
elastic net (causally tuned)†	866	0.008	0.112	-1.02	0.54	-2.48	17.72	2.16	1.16	265	30.6	-3.40

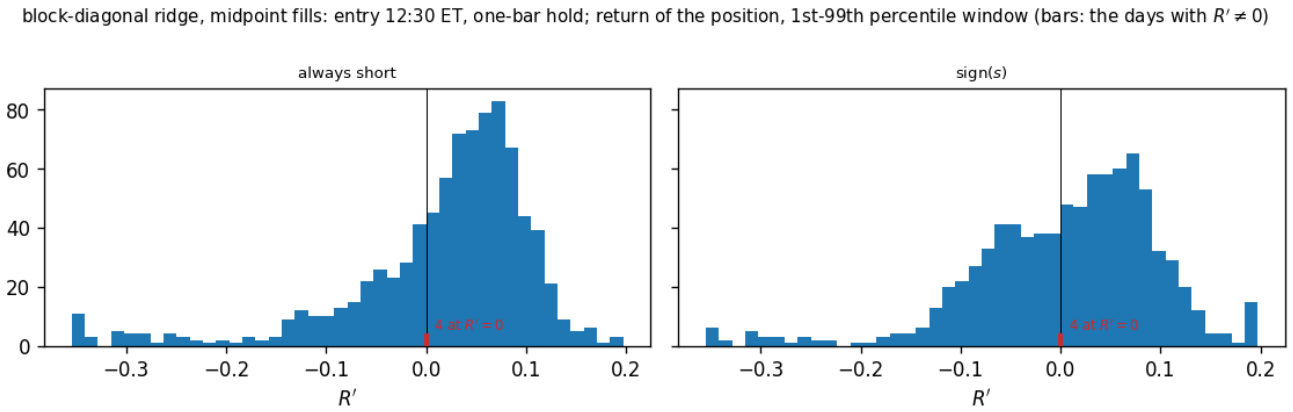


Distribution of the return of the position taken at 12:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 12:30 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.013	0.115	-1.23	0.23	-3.74	25.06	3.29	1.78	0	0.0	-2.88
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.006	0.116	-1.23	0.68	-2.53	23.49	1.59	0.86	252	29.1	-3.78
block-diagonal ridge	866	0.006	0.116	-1.23	0.68	-2.15	23.38	1.53	0.82	278	32.1	-3.85
block-diagonal ridge, without the FOMC columns	866	0.006	0.116	-1.23	0.68	-2.11	23.38	1.54	0.83	285	32.9	-3.83
LightGBM†	866	0.007	0.116	-1.23	0.62	-2.58	23.54	1.66	0.90	268	30.9	-3.76
XGBoost†	866	0.009	0.116	-1.23	0.62	-2.65	23.86	2.20	1.19	259	29.9	-3.48
lasso (causally tuned)†	866	0.006	0.116	-1.23	0.68	-2.14	23.34	1.46	0.79	269	31.1	-3.87
lasso ($\alpha = 10^{-4}$)	866	0.006	0.116	-1.23	0.68	-2.15	23.41	1.60	0.86	277	32.0	-3.81
elastic net (causally tuned)†	866	0.003	0.116	-1.23	0.68	-2.19	23.10	0.86	0.47	262	30.3	-4.19

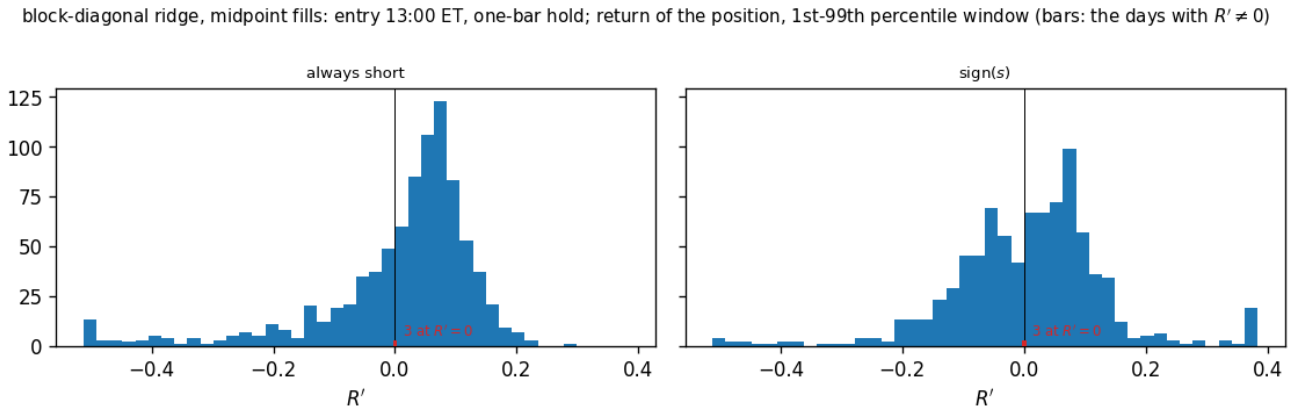


Distribution of the return of the position taken at 12:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 12:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.002	0.168	-1.72	0.29	-4.23	29.08	0.43	0.23	0	0.0	-3.12
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.008	0.168	-1.72	1.30	-0.99	29.14	1.41	0.76	335	38.7	-2.63
block-diagonal ridge	866	0.012	0.168	-1.72	1.49	0.63	28.93	2.09	1.13	334	38.6	-2.27
block-diagonal ridge, without the FOMC columns	866	0.015	0.168	-1.72	1.49	0.64	29.03	2.56	1.38	341	39.4	-2.03
LightGBM†	866	0.017	0.167	-1.72	1.49	0.67	29.13	2.93	1.58	341	39.4	-1.83
XGBoost†	866	0.016	0.168	-1.72	1.49	0.64	29.09	2.78	1.50	339	39.1	-1.91
lasso (causally tuned)†	866	0.009	0.168	-1.72	1.30	-0.94	29.17	1.52	0.82	323	37.3	-2.57
lasso ($\alpha = 10^{-4}$)	866	0.012	0.168	-1.72	1.49	0.62	28.93	2.06	1.11	330	38.1	-2.29
elastic net (causally tuned)†	866	0.007	0.168	-1.72	1.30	-1.03	29.11	1.28	0.69	319	36.8	-2.70

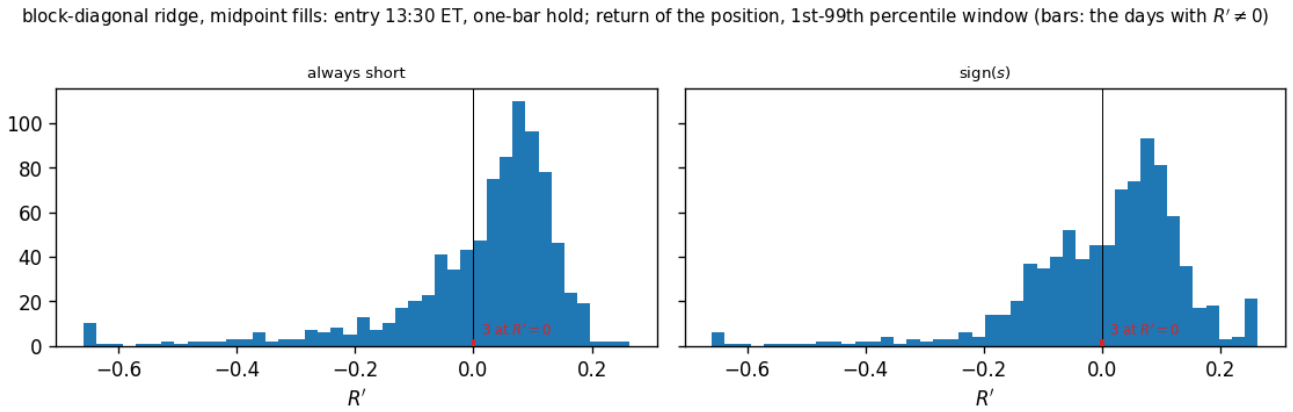


Distribution of the return of the position taken at 13:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 13:30 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.008	0.161	-1.07	0.25	-2.85	11.71	1.46	0.79	0	0.0	-3.47
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.009	0.161	-1.07	0.96	-1.25	11.46	1.71	0.92	189	21.8	-3.40
block-diagonal ridge	866	0.008	0.161	-1.07	0.96	-1.40	11.44	1.51	0.82	205	23.7	-3.52
block-diagonal ridge, without the FOMC columns	866	0.009	0.161	-1.01	1.07	-0.75	11.34	1.71	0.92	215	24.8	-3.42
LightGBM†	866	0.005	0.161	-1.07	0.85	-2.01	11.34	0.85	0.46	193	22.3	-3.82
XGBoost†	866	0.006	0.161	-1.07	0.85	-1.90	11.38	1.02	0.55	197	22.7	-3.73
lasso (causally tuned)†	866	0.005	0.161	-1.07	0.96	-1.38	11.28	0.89	0.48	200	23.1	-3.83
lasso ($\alpha = 10^{-4}$)	866	0.008	0.161	-1.07	0.96	-1.40	11.42	1.44	0.78	205	23.7	-3.56
elastic net (causally tuned)†	866	0.005	0.161	-1.07	0.96	-1.45	11.29	0.91	0.49	194	22.4	-3.81

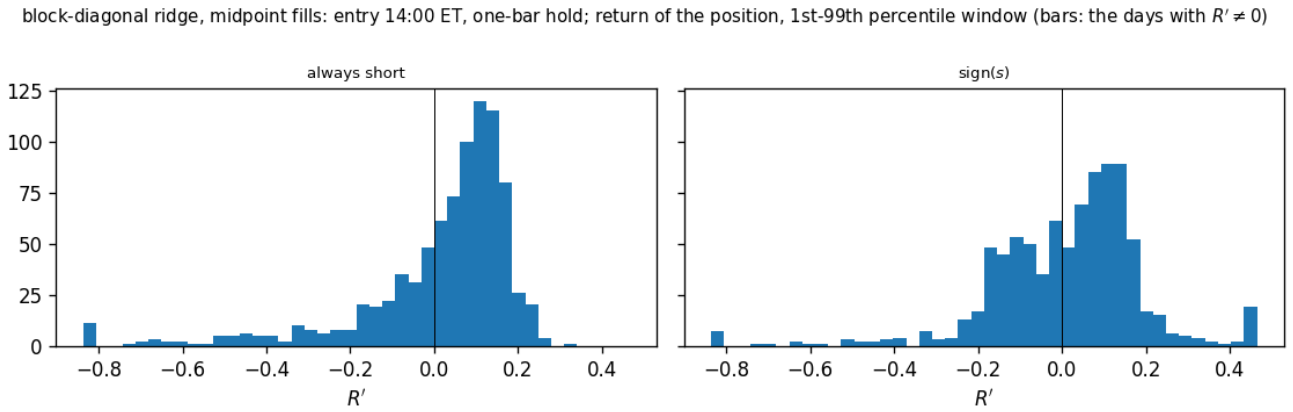


Distribution of the return of the position taken at 13:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 13:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.008	0.237	-2.65	0.32	-4.48	32.87	0.93	0.50	0	0.0	-2.62
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.011	0.237	-2.65	1.63	-1.89	32.73	1.35	0.73	335	38.7	-2.43
block-diagonal ridge	866	0.011	0.237	-2.65	1.63	-2.62	32.85	1.34	0.72	325	37.5	-2.44
block-diagonal ridge, without the FOMC columns	866	0.011	0.237	-2.65	1.63	-2.41	32.82	1.35	0.73	333	38.5	-2.43
LightGBM†	866	0.007	0.237	-2.65	1.63	-1.85	32.51	0.86	0.46	350	40.4	-2.68
XGBoost†	866	0.006	0.237	-2.65	1.63	-1.83	32.46	0.73	0.39	360	41.6	-2.76
lasso (causally tuned)†	866	0.009	0.237	-2.65	1.63	-2.39	32.69	1.09	0.59	337	38.9	-2.57
lasso ($\alpha = 10^{-4}$)	866	0.008	0.237	-2.65	1.63	-2.58	32.67	1.00	0.54	325	37.5	-2.62
elastic net (causally tuned)†	866	0.008	0.237	-2.65	1.63	-2.40	32.63	0.98	0.53	334	38.6	-2.63

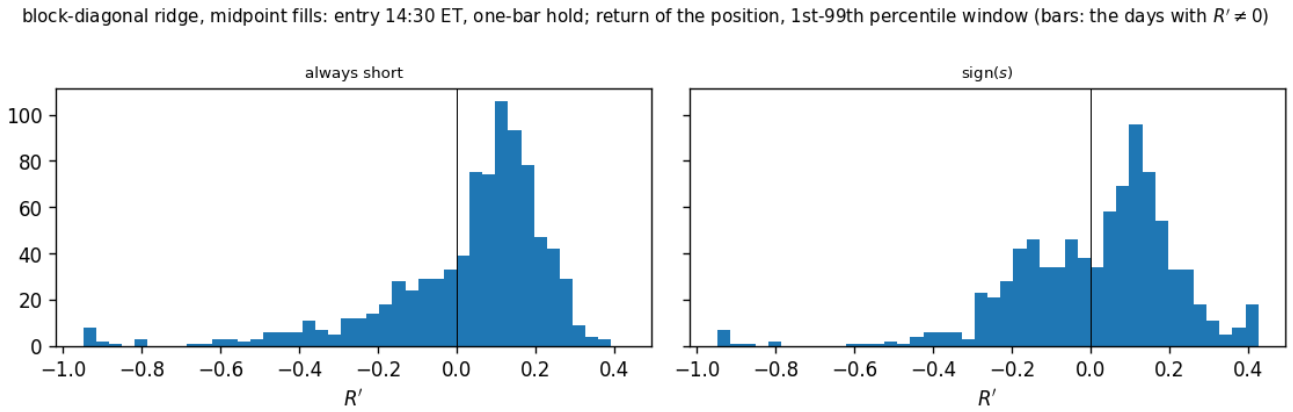


Distribution of the return of the position taken at 14:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 14:30 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.019	0.300	-5.15	0.38	-7.56	108.09	1.86	1.00	0	0.0	-1.52
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.012	0.300	-5.15	2.01	-5.59	106.58	1.20	0.64	232	26.8	-1.90
block-diagonal ridge	866	0.011	0.300	-5.15	1.91	-6.57	106.61	1.10	0.59	243	28.1	-1.94
block-diagonal ridge, without the FOMC columns	866	0.007	0.300	-5.15	1.91	-6.54	106.07	0.70	0.38	250	28.9	-2.16
LightGBM†	866	0.009	0.300	-5.15	1.91	-6.21	106.22	0.85	0.46	263	30.4	-2.08
XGBoost†	866	0.005	0.300	-5.15	1.91	-6.19	105.82	0.51	0.28	263	30.4	-2.27
lasso (causally tuned)†	866	0.011	0.300	-5.15	1.91	-6.28	106.55	1.09	0.59	252	29.1	-1.96
lasso ($\alpha = 10^{-4}$)	866	0.008	0.300	-5.15	1.91	-6.54	106.21	0.81	0.44	246	28.4	-2.10
elastic net (causally tuned)†	866	0.013	0.300	-5.15	1.91	-6.27	106.84	1.30	0.70	250	28.9	-1.84

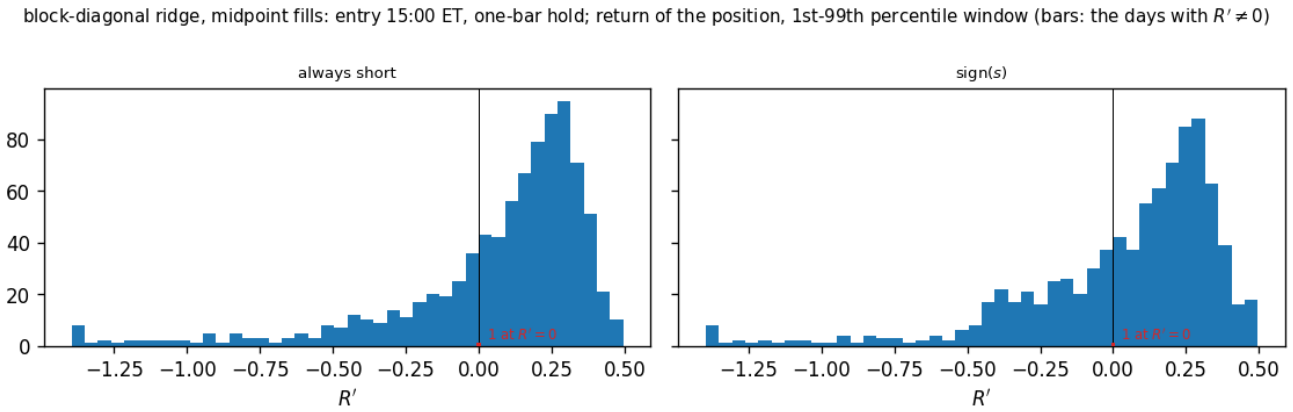


Distribution of the return of the position taken at 14:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 14:30 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:00 ET, one-bar hold

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.060	0.379	-3.31	0.58	-3.03	15.14	4.64	2.51	0	0.0	0.18
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.047	0.381	-3.31	2.85	-1.80	13.85	3.60	1.94	105	12.1	-0.36
block-diagonal ridge	866	0.037	0.382	-3.31	1.15	-2.67	13.88	2.89	1.56	108	12.5	-0.73
block-diagonal ridge, without the FOMC columns	866	0.037	0.382	-3.31	1.15	-2.66	13.85	2.83	1.52	110	12.7	-0.76
LightGBM†	866	0.042	0.382	-3.31	1.29	-2.62	14.04	3.22	1.74	110	12.7	-0.56
XGBoost†	866	0.048	0.381	-3.31	2.85	-1.72	13.85	3.68	1.98	115	13.3	-0.32
lasso (causally tuned)†	866	0.042	0.382	-3.31	1.29	-2.61	14.07	3.27	1.76	111	12.8	-0.53
lasso ($\alpha = 10^{-4}$)	866	0.039	0.382	-3.31	1.15	-2.67	13.93	2.97	1.60	106	12.2	-0.69
elastic net (causally tuned)†	866	0.040	0.382	-3.31	1.29	-2.59	13.97	3.11	1.68	115	13.3	-0.61

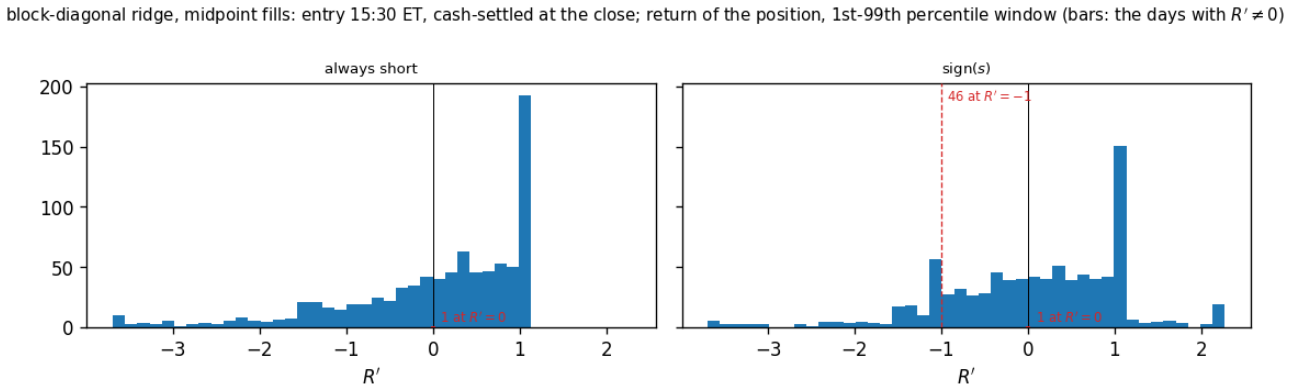


Distribution of the return of the position taken at 15:00 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).

One trade a day: enter the nearest-OTM straddle at 15:00 ET and exit thirty minutes later at the next stamp, in the same two strikes. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — entry 15:30 ET, cash-settled at the close

	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{clk}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{clk}}$
Short volatility (always short)												
all models (no forecast)	866	0.014	1.128	-10.32	1.00	-2.31	10.89	0.38	0.20	0	0.0	-0.26
sign(s) (± 1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.030	1.127	-10.32	6.84	-1.38	10.93	0.79	0.43	148	17.1	-0.04
block-diagonal ridge	866	0.091	1.124	-5.42	10.32	0.60	10.71	2.38	1.28	234	27.0	0.81
block-diagonal ridge, without the FOMC columns	866	0.096	1.124	-5.42	10.32	0.59	10.72	2.52	1.36	236	27.3	0.89
LightGBM†	866	0.101	1.123	-5.42	10.32	0.42	10.79	2.64	1.43	165	19.1	0.96
XGBoost†	866	0.090	1.124	-10.32	6.84	-1.36	11.34	2.34	1.26	162	18.7	0.80
lasso (causally tuned)†	866	0.106	1.123	-6.84	10.32	-0.02	10.97	2.79	1.50	184	21.2	1.03
lasso ($\alpha = 10^{-4}$)	866	0.075	1.125	-6.84	10.32	0.11	10.83	1.96	1.06	246	28.4	0.59
elastic net (causally tuned)†	866	0.057	1.126	-6.84	10.32	0.01	10.82	1.49	0.81	182	21.0	0.34

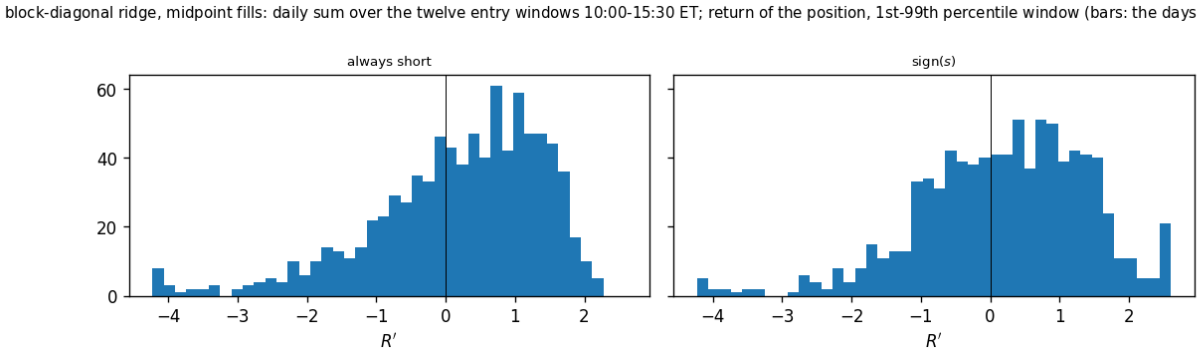


Distribution of the return of the position taken at 15:30 ET, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window; the mass at -1 is the days the straddle expires worthless).

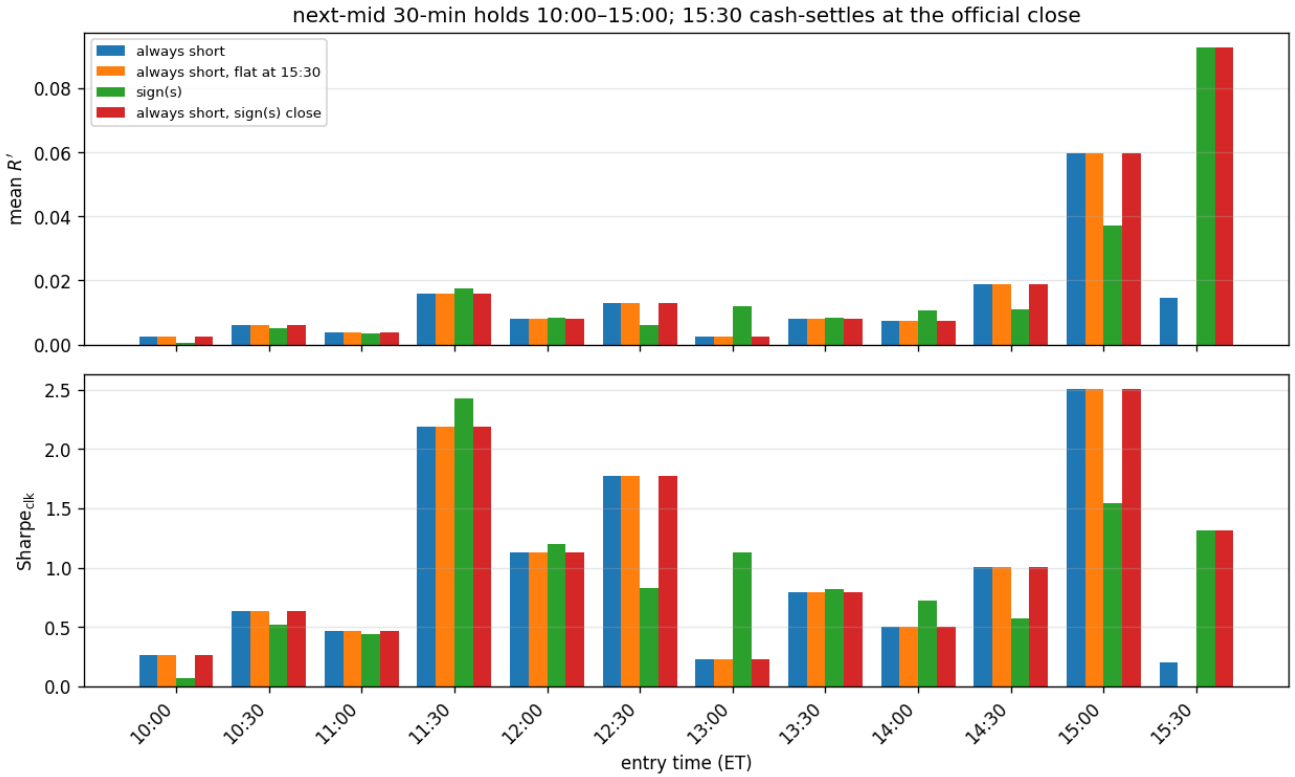
One trade a day: enter the nearest-OTM straddle at 15:30 ET and let it cash-settle at the official close. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Rule table by strategy — pooled over the twelve entry windows

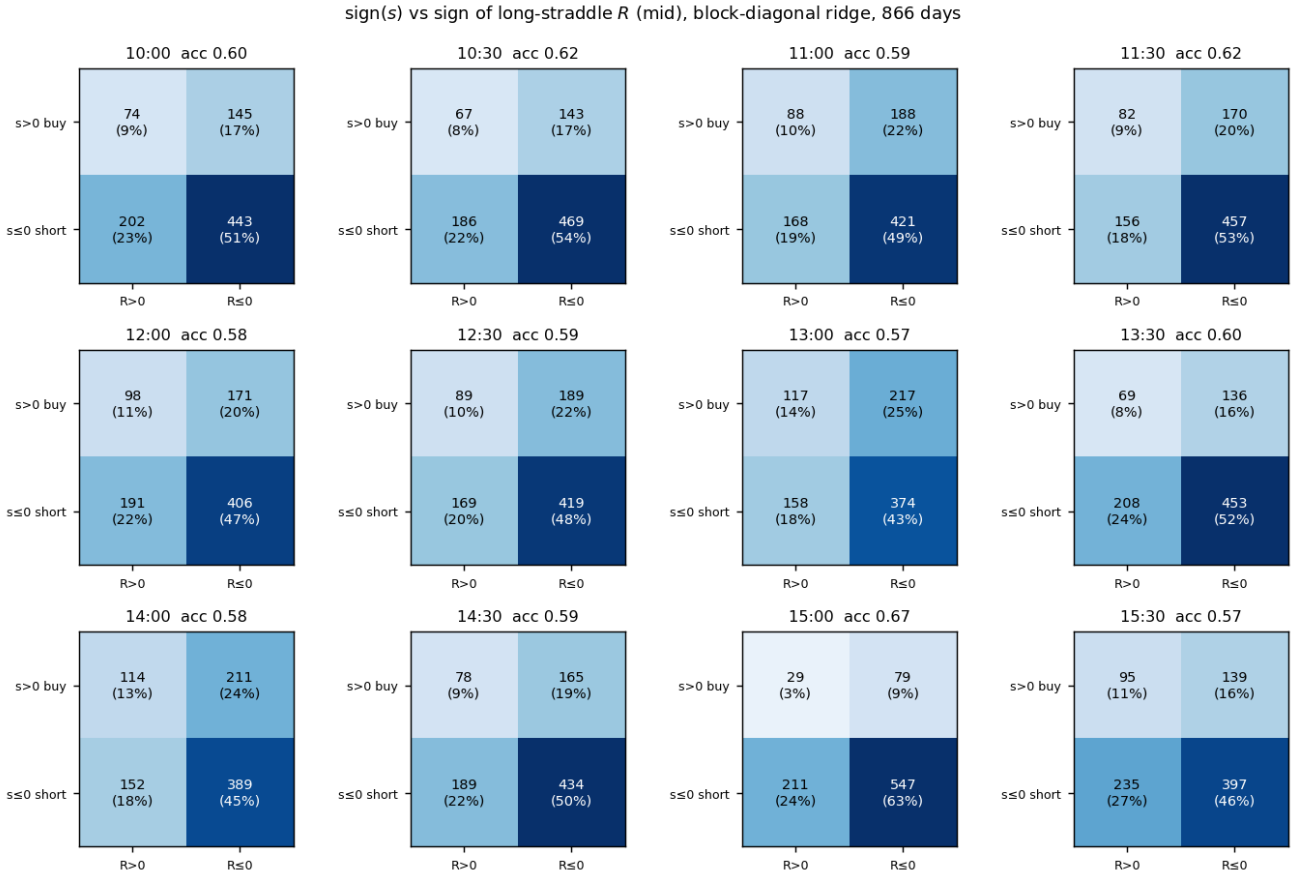
	n	mean	std	min	max	skew	ex. kurt	t	$\text{Sharpe}_{\text{ann}}$	n_{buy}	%buy	$\text{Sharpe}^{\times}_{\text{ann}}$
Short volatility (always short)												
all models (no forecast)	866	0.160	1.338	-11.40	2.24	-1.92	8.73	3.53	1.90	0	0.0	-1.92
sign(s) (±1 *on the sign of* s)												
baseline (HAR + calendar OLS)	866	0.161	1.321	-8.94	6.22	-1.00	4.63	3.58	1.93	122	14.1	-2.36
block-diagonal ridge	866	0.212	1.320	-6.15	10.65	0.16	6.23	4.72	2.55	149	17.2	-1.72
block-diagonal ridge, without the FOMC columns	866	0.219	1.321	-6.15	10.65	0.15	6.25	4.88	2.63	157	18.1	-1.64
LightGBM†	866	0.223	1.296	-6.23	10.65	0.00	6.35	5.07	2.73	122	14.1	-1.62
XGBoost†	866	0.216	1.312	-9.98	6.95	-0.93	6.36	4.84	2.61	119	13.7	-1.71
lasso (causally tuned)†	866	0.216	1.346	-7.45	11.70	-0.14	8.51	4.73	2.55	130	15.0	-1.64
lasso ($\alpha = 10^{-4}$)	866	0.193	1.310	-6.72	10.10	-0.22	5.62	4.33	2.34	157	18.1	-1.94
elastic net (causally tuned)†	866	0.169	1.312	-7.45	11.70	-0.05	9.38	3.78	2.04	137	15.8	-2.23



Distribution of the return of the daily sum over the twelve entry windows, block-diagonal ridge forecast, one panel per rule (returns inside their 1st–99th percentile window).



Mean R' and $\text{Sharpe}_{\text{clk}}$ by entry clock (grouped bars). $\text{Sharpe}_{\text{clk}}$ is that clock's daily series $\times\sqrt{252}$, not the pooled $\text{Sharpe}_{\text{ann}}$. Next-mid holds 10:00–15:00 (25 minutes from 09:35, 30 minutes from every later clock); 15:30 cash-settles at the official close.



Confusion matrix per entry clock: sign(s) (buy if $s > 0$) against the sign of the long-straddle midpoint return. 15:30 is the last-30-min trade.

Each row is the daily sum over the twelve entry windows 10:00–15:30 ET: one number per expiration day, then the usual statistics on that daily series. 866 expiration days, 2020-01-03 to 2024-04-30 (the deck's frame). Fills are at the quoted midpoint everywhere except the last column, which is the same rule filled at the crossed spread: entry at the touch (the ask when long, the bid when short) and exit at the touch of the next stamp; the 15:30 leg cash-settles and pays no exit spread, and a re-pick that lands on the same two strikes with the same sign is a hold, not a round trip, so it is not charged.

Notes

Straddle here means the nearest out-of-the-money call plus the nearest out-of-the-money put, same-day expiry, bought or sold as one position. One expiration day = one return; mid fill. Every t here is the plain $t = \sqrt{n} \cdot \text{mean/std}$, with no autocorrelation correction. "ex. kurt" is the bias-corrected sample excess kurtosis (Gaussian = 0). The short-volatility rule takes no forecast, so it is one row for all models. **Two Sharpes.** Both multiply by $\sqrt{252}$ because an expiration day is the unit of time (same year-length as the paper). They are not the same object. On a *single-clock* page, $\text{Sharpe}_{\text{clk}} = (\text{mean/std}) \times \sqrt{252}$ of that clock's series: one 30-minute hold per expiration day. It answers "if I only entered at this clock, what is my annual Sharpe?" It is *not* the book's $\text{Sharpe}_{\text{ann}}$. On the *pooled* page, $\text{Sharpe}_{\text{ann}} = (\text{mean/std}) \times \sqrt{252}$ of $R_d^{day} = \sum_t R'_{d,t}$, the sum of the day's twelve holds. That reserved name is the book's daily P&L after pooling the session. At 15:30 the two coincide (one trade that day); they do not on 10:00–15:00 or on the pooled page. $\text{Sharpe}^{\times}$ is the same convention at the crossed spread. Every other column is daily. n_{buy} is the number of expiration days with position > 0 (always-short is 0; on the pooled page it is days the sum of the twelve positions is positive); %buy is $100 \cdot n_{\text{buy}}/n$, the notebook's column. The other column this table adds to the deck's is $\text{Sharpe}^{\times}_{\text{ann}}$, the same rule's annualized Sharpe at the *crossed spread* instead of the midpoint; every other column is the deck's.

Forecasts: baseline (HAR + calendar OLS); block-diagonal ridge (HAR block and exogenous block, separate penalties) on the design carrying the FOMC calendar block, and the same ridge on the earlier design without those columns, as a diagnostic row; LightGBM and XGBoost on the all-features design; lasso on the same design, causally tuned vs. fixed $\alpha = 10^{-4}$; elastic net (causally tuned). † marks a column still fitted on the earlier design panel; a run of those four on the panel of record is pending. The signal is $s_t = \widehat{RV}_t - \text{IV}^2_{\text{hr}} h_t w_t$: the forecast against the implied variance of the same window. w_t is the expanding mean of each prior day's remaining-session share of realized variance, $\text{RV}_t / \sum_{s \geq t} \text{RV}_s$ (in-fit panel history back to 2001), not the ratio of trailing clock-means. On the bars whose vendor implied volatility is a censored solver node the deck's treatment is used here too — the volatility that reproduces the straddle midpoint, recovered by bisection over the remaining h_t hours — so every bar carries a signal and no bar sits flat.